

MASS-ENG-002

Enterprise Core Specification

Page 1 – Purpose & Scope

Purpose

The Enterprise Core provides the shared infrastructure for MASS HQ. All platform engines depend on this subsystem.

Objectives

- Establish enterprise boundaries
- Provide shared configuration
- Register modules and services
- Coordinate cross-module communication
- Publish enterprise events

Out of Scope

Business logic, AI reasoning, and domain-specific workflows.

Page 2 – Architecture

Core Components

Enterprise Core contains: Configuration Service, Module Registry, Service Registry, Event Bus Interface, Health Monitoring, Startup Coordinator.

Relationships

All engines consume Enterprise Core; Enterprise Core consumes no business engines.

Page 3 – Functional Specification

Requirements

1. Maintain a single enterprise configuration.
2. Register modules during startup.
3. Expose service discovery.
4. Publish enterprise events.
5. Expose health status.

Non-Functional

Deterministic startup, extensibility, configuration-driven behavior.

Page 4 – Interfaces & Examples

Interfaces

Module Registration: input (module metadata), output (registration status).

Configuration Service: input (key), output (value).

Event Publisher: input (event, payload), output (event id/status).

Example Startup

Core → Configuration → Registry → Event Bus → Identity → Security → Workflow → Ready.

Page 5 – Construction Package

Claude Checklist

Create Enterprise Core module; implement configuration service; module registry; service discovery; event publisher; health monitoring; startup coordinator; tests.

Definition of Done

Modules register, configuration loads, events publish, services discoverable, health endpoints

operational.

Constitution References

Enterprise Governance, Enterprise Structure, Identity, Security, Platform Principles.